UTER SCIENCE & ENGINEERING
ERSITY COLLEGE OF ENGINEERING (A)
Osmania University, Hyderabad-07

CLASS TEST-I

Subject : Data Analytics using R

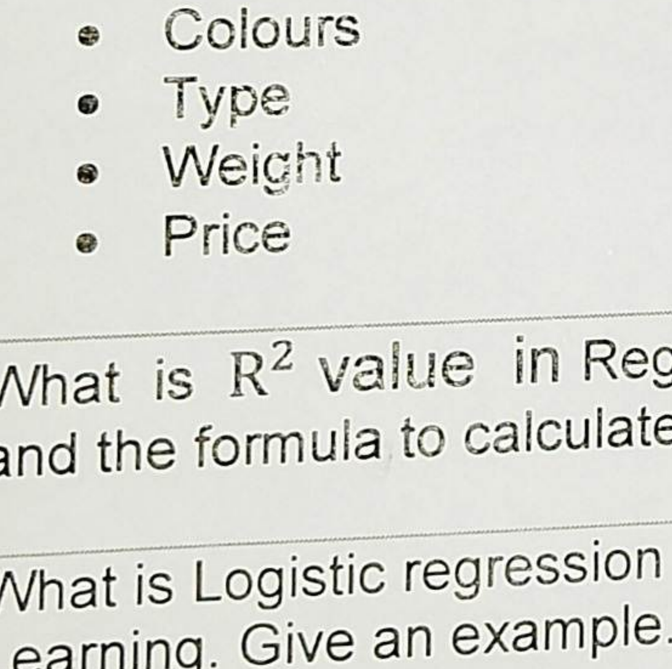
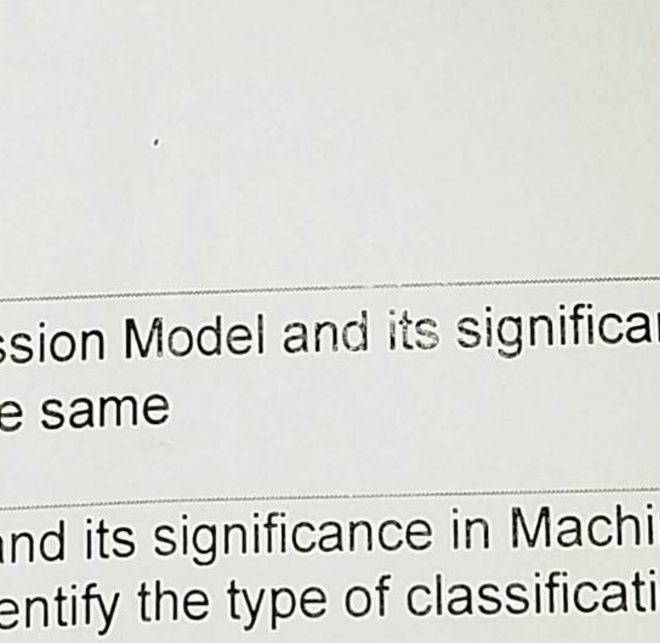
Class: B.E. (CSE)
Semester: IV
Academic Year: 2024-2025

Max Marks: 30
Duration:1.5 hr(2:30 p.m. to 4:00 p.m.)
Date : 29-03-2025

Note: If any data missing may be assumed
Answer all questions in Part A

Q No	PART A Answer All Questions (5 x 2 = 10 Marks)	Marks	BT	CO
1.	What is the purpose of the function "summary" in R programming Language? List out the outputs of function "summary"	2	2	1

2.	Find out what will be the output for the following program if the **num** value is : **100, -100, 20, 9**
	`if (num > 10) {`
	`  print("The number is positive")`
	`} else if (num < 0) {`
	`  print("The number is negative")`
	`} else {`
	`  print("The number is Zero other")`
	`}`
		2	4	1

3.	What is data frame in R Program? Write a R program for creating a data frame for fruits with sub fields of fruits • Colours • Type • Weight • Price	2	3	1
4.	What is R^2 value in Regression Model and its significance and the formula to calculate the same	2	1	2
5.	What is Logistic regression ? and its significance in Machine Learning. Give an example. Identify the type of classification for the Fig 1 and Fig 2  Fig 1  Fig 2	2	5	2

PART B

Answer Any Two Questions (2 x 10 = 20Marks)

a)	List out what are functions for reading the data from various sources like file, table, keyboard etc. Write the syntax for the same	6	3	1
b)	Write a program in R to represent following data in a matrix as per below instructions			
	i) Matrix in row wise and print the output Data: 10, 2, 24, 17, 9, 11, 15, 19, 25			
	ii) Matrix having 3 columns and the print Data: 1,2, 3,4,5,6,7,8,9,10, 12			
a)	Following is the data of Iris flowers. What will be the output of the following program if we create this as a data set in and run the following program data<- iris print(data) head(data) summary(data$Petal.Width)- mention the parameters (no need to run R program)	7		

	Sepal.Length	Sepal.Width	Petal.Length	Petal.Width
1	5.1	3.5	1.4	0.2
2	4.9	3.0	1.4	0.2
3	4.7	3.2	1.3	0.2
4	4.6	3.1	1.5	0.2
5	5.0	3.6	1.4	0.2
6	5.4	3.9	1.7	0.4
7	4.6	3.4	1.4	0.3
8	5.0	3.4	1.5	0.2
9	4.4	2.9	1.4	0.2
10	4.9	3.1	1.5	0.1

b)	Write a R program to generate a scatter plot for the following given data: • x = c(1, 2, 3, 4, 5, 6, 7, 8, 9, 10) • y = c(10, 9, 8, 7, 6, 5, 4, 3, 2, 1) Label the x-axis as " X Data" and the y-axis as "Y Data", and give the title "Class Test 1-Scatter Plot ".	5	3	1
a)	Following is the data of a selected group of people. Please write a R program to find out the regression model and also to print the graph.	6	4	2

Age	Weight
10	40
20	50
30	60
40	70
80	110

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b)	What is Decision Tree in Machine learning? and its significance? List out 3 Decision tree algorithms used in Machine Learning	4	2	2